

Long-term durability prediction models for GFRP bars in concrete environment

Julio F Davalos¹, Yi Chen² and Indrajit Ray¹

Abstract

This article presents a durability study of glass fiber reinforced polymer (GFRP) bars within concrete environment and provides recommendations for effective accelerated ageing methods, data reduction and validation, and formulation of master curves for long-term predictions. Concrete beams reinforced with GFRP bars were immersed in tap water inside temperature-controlled tanks. Sustained load was applied to beam samples, and elevated temperatures were used to accelerate the ageing of GFRP bars. After exposure, GFRP bars were tested for residual tensile strength, which was considered as a measure of the durability performance of the specimens. Scanning electron microscopy (SEM) images were used to evaluate the microstructural changes of conditioned specimens. Short-term (up to 270 days) results showed that compared to alkalinity and low sustained load, moisture and elevated temperature played much more important roles in accelerating environmental attack to GFRP bars. Based on the short-term test results, the dominant degradation mechanism for GFRP bars in concrete was found as the deterioration of fiber/matrix interface. A degradation model based on Arrhenius concept and prediction procedures are given, and correlations with other studies and field data are provided. The tensile strength retentions for GFRP bars in saturated concrete subjected to sustained load were predicted. It was found that degradation rates decreased rapidly, but tensile strength retentions at different exposure temperatures converge to a constant value with the increase of exposure time. Beyond fiber-matrix debonding, there is a possibility that other degradation mechanisms such as fiber corrosion may start to dominate the deterioration process of GFRP bars with time.

Keywords

Ageing, bars, durability, fiber reinforced polymers, glass fiber, prediction model

Introduction

There is pressing concern with ageing and deterioration of infrastructure, particularly of reinforced concrete structures, due to corrosion of steel reinforcement. To mitigate this problem, epoxy-coated steel bars are being used, but field and laboratory data have shown inadequate protection offered by this technology.¹ Thus, FRP bars have been developed as a possible long-term solution to overcome the corrosion of steel in concrete structures. Due to their low cost, glass fiber reinforced polymer (GFRP) bars are used more extensively than carbon FRP bars. But glass fibers and the commonly used thermoset resin matrices are known to be susceptible to attack by moisture, elevated temperature, and alkalinity. The degradation of GFRP bars may result from polymer degradation, fibre-matrix debonding, and fiber-stress corrosion.^{2–4} Moreover, short-term results from most durability studies^{2–5}

showed that GFRP bars experienced significant loss in mechanical properties, especially tensile strength. Although recently ISIS Canada⁶ reported that no degradation was found by using analytical methods (e.g. scanning electron microscopy; SEM) for GFRP bars extracted from several field structures after 5 to 8 years of service, there is a general lack of long-term performance data for GFRP bars in civil infrastructure. When considering 50 to 75 years of expected life cycle for civil infrastructure, exposure to harsh

¹Department of Civil and Environmental Engineering, Center for Transportation Security and Infrastructure Innovations (TransI²), West Virginia University, Morgantown, USA

²SWSG PC, Reston VA, USA

Corresponding author:

Yi Chen, SWSG PC, Reston VA, 20190
Email: sjtuchenyi@yahoo.com

environmental conditions can be anticipated for GFRP bars. As a result, design guidelines^{7,8} for FRP-reinforced concrete structures recommend high safety factors and low sustained stress levels, or even exclude the use of GFRP bars in many applications. But these limits are usually given without support of extensive experimental data. These conservative prescriptions preclude taking full advantage of superior properties of FRP, such as high strength to weight ratio, good resistance to attack by chloride ions, and electromagnetic neutrality. Moreover, overly conservative prescriptions can hinder the wider acceptance of GFRP bars in civil engineering. Therefore, there is a need for development of effective test methods and prediction models by which long-term durability performance of GFRP bars in concrete environment can be assessed.

Prediction models

Numerous studies have been carried out on the durability of GFRP bars, but few tests were conducted for FRP bars within actual concrete,^{2,9-12} as opposed to simulated concrete pore-solutions. Moreover, only few studies were directed to the development of life-cycle durability prediction models for FRP bars in concrete environment. In general, there are four typical degradation models for FRP bars. The durability prediction procedures for all those models were based on Arrhenius equation as shown in Equation (1):

$$k = A \exp\left(\frac{-E_a}{RT}\right) \quad (1)$$

where k = degradation rate (1/time); A = constant related to material and degradation process; E_a = activation energy; R = universal gas constant; and T = temperature in Kelvin. The primary assumption of predictions using Arrhenius relation is that the single dominant degradation mechanism will not change with time and temperature during the exposure period, but the degradation rate will be accelerated with the increase of exposure temperature. According to Equation (1), for two different exposure temperatures, the acceleration factor for a range of high temperature (T_1) to low temperature (T_0) can be calculated as:

$$\begin{aligned} AF = \frac{t_0}{t_1} &= \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{A \exp\left(\frac{-E_a}{RT_1}\right)}{A \exp\left(\frac{-E_a}{RT_0}\right)} \\ &= \exp\left[\frac{E_a}{R} \left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \end{aligned} \quad (2)$$

where, AF is the acceleration factor; t_1 and t_0 are the times required for some property to reach a given

retention value at temperatures of T_1 and T_0 , respectively; c is a constant; k_1 and k_0 are the degradation rates at temperatures of T_1 and T_0 , respectively. Equation (1) can also be transformed into:

$$\ln\left(\frac{1}{k}\right) = \frac{E_a}{R} \frac{1}{T} - \ln(A) \quad (3)$$

From Equation (2), k can be expressed as the inverse of time needed for some property to reach a given retention value. From Equation (3) one can further observe that the logarithm of time needed for some property to reach a given retention value is a linear function of $1/T$ with the slope of E_a/R . According to Equation (3), to use the prediction procedure based on Arrhenius relation, one should obtain straight (for different exposure temperatures) and parallel (for different retention values) lines in Arrhenius plots, which will be shown later when the durability prediction is carried out using the test results of this study. The typical degradation models used for FRP bars will be discussed next.

The first model type was presented by Bank et al.¹³ The residual strength and time was assumed as follows:

$$Y = a \log(t) + b \quad (4)$$

where, in this and all other models presented in this article Y is the property retention (%), and t is the exposure time; while a and b are regression constants. This model was first developed for the residual strength of glass fiber concrete (GRC) and was successfully applied to durability prediction with data of 10 years.¹⁴ It is also the most widely used degradation model for FRP bars. The time shift factor prediction method developed by DeJke² and the prediction procedure using the time-temperature superposition proposed by Caceres et al.¹⁵ can both be considered to follow this model type. There are, however, intrinsic limitations with this model. First, it is only a phenomenological representation of test data and there is no hypothesis concerning the degradation mechanism. Second, according to the assumed relationship of Equation (4), the strength approaches infinity at time zero. When this model was applied to GFRP bar test results for durability prediction based on Arrhenius relation, Gonenc³ found that the 'Arrhenius plots' were not parallel to each other, which indicated that the degradation mechanism had changed with exposure time, violating the fundamental assumption in the Arrhenius relation of maintaining consistent degradation mechanism.

The second type of model was also first developed for GRC,¹⁶ with the degradation mechanism assumed to be stress corrosion of glass fibers. A static fatigue

model was defined to describe the residual strength and exposure time relationship:

$$Y = 1/\sqrt{1 + gt} \quad (5)$$

where g is a function of temperature, stress, and solution concentration as given by:

$$g = g_T g_0 \exp\left(-\frac{E_a}{RT}\right) \quad (6)$$

where, g_T is a function of temperature, g_0 accounts for the pH of environment and initial state of fibers. The difficulty with this model is that as shown in Equation (6), g_T is a function of exposure temperature. Thus, for example, when the activation energy was calculated by Beddow et al.¹⁶ using Equation (7) below, the effect of g_T was ignored without explanation (i.e. the last term on the right side of Equation 7 was ignored):

$$\ln(g_2) = \ln(g_1) - \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) + \ln\left(\frac{g_{T_1}}{g_{T_2}}\right) \quad (7)$$

where g_1 , g_2 , g_{T_1} , and g_{T_2} are values corresponding, respectively, to the temperatures of T_1 and T_2 .

The third model type was derived by the present authors from the following basic equation:

$$f = \beta t^\alpha \quad (8)$$

where f is the rate of change, β is the rate constant, and α is a material constant. Equation (8) is a common expression used to describe rate of changes in material properties with time¹⁷ and allows including models based on moisture absorption. To formulate a prediction equation, it is assumed that the ultimate tensile load capacity of GFRP bars is proportional to remaining unaffected cross-sectional area after conditioning. From Equation 8, the tensile retention capacity (%) of bars vs. time can be expressed as:

$$\begin{aligned} Y &= 100 \left(1 - \frac{\int_0^t f dt}{r_0} \right)^2 = 100 \left(1 - \frac{\beta t^{\alpha+1}}{(\alpha+1)r_0} \right)^2 \\ &= 100(1 - jt^{\alpha+1})^2 \end{aligned} \quad (9)$$

where r_0 is the initial radius of the cross-section; and j is a factor accounting for temperature, solution concentration, and other experimental conditions. For $\alpha = -0.5$, Equation (9) becomes equivalent to prediction models based on moisture absorption,¹⁸ as

$$Y = 100 \left(1 - \frac{\sqrt{2DCt}}{r_0} \right)^2 \quad (10)$$

where D is the diffusion coefficient and C is the concentration of solution. In this case, diffusion models based on Fick's law are employed to correlate the retention in tensile strength with moisture absorption of FRP bars. But the difficult task of conducting moisture absorption tests can be avoided by using instead Equation (9), which can also be applied to specimens exposed to water since the concentration of solution is not required as is in Equation (10). One problem with these models is the assumption of discounting completely the affected area of FRP bars; that is the area with diffused chemicals may not be completely degraded and still able to carry some load.

The fourth model type was first developed using acousto-ultrasonic technique to predict flexural strength retention of composite laminates.¹⁹ The degradation mechanism for this model is assumed to be debonding at the fiber/matrix interface, with the following formulation:

$$Y = (100 - Y_\infty) \exp(-t/\tau) + Y_\infty \quad (11)$$

where, Y_∞ is the tensile strength retention (%) at exposure time of infinity and τ is a characteristic time dependent on temperature. Details are given in Ref 4, where following this Equation, a prediction procedure was developed by the present authors for tensile strength retention of GFRP bars in simulated concrete pore solutions, and an Arrhenius analysis was also included in the prediction procedure to check the validity of accelerated ageing tests.

Objective

Based on above discussion, the degradation model and prediction procedures of all four model types were previously developed from or successfully applied to test results of bare FRP bars exposed to simulated solutions, for several types of FRP bars, and exposure conditions. At present, there are no prediction models based on results for GFRP bars embedded in concrete and under sustained loading. Thus, in this article, the durability of GFRP bars is investigated for samples exposed to actual concrete environment and subjected to sustained loading. Based on test results, durability predictions are formulated and evaluated for long-term axial strength retention, and recommendations for applications in practice are given. Based on a comprehensive study for accelerated ageing test environments for GFRP bars,²⁰ the critical environment of exposure to natural alkaline solution was selected by immersing in tap water FRP bars embedded in concrete specimens and using elevated temperature as the accelerating agent. This procedure was successfully used for short-term accelerated tests of bare GFRP bars in direct

alkaline solutions, leading to long-term behavior prediction by Arrhenius procedure.⁴

Experimental program

Specimens

In this study, two types of commercially available GFRP bars with 9.53 mm diameter (No. 3) were used. GFRP bars made of E-glass fibers and vinyl ester resin were selected due to their wide application in civil engineering infrastructure. Both of them had a glass fiber content of above 70% by weight and were manufactured by the pultrusion process. As shown in Figure 1, GFRP1 bar was helically wrapped and slightly surface sand-coated, and GFRP2 bar was fully surface sand-coated.

In order to investigate the effect of concrete environment on GFRP bars, concrete beams reinforced with GFRP bars were cast with cross section of $0.1\text{ m} \times 0.1\text{ m}$ and length of 0.9 m . The concrete mix design was cement 380, sand 730, Gravel 1141, and water 171 kg/m^3 . And the 28-day compressive strength was about 53 MPa. The concrete beams not subjected to sustained load are shown in Figure 2(a). As shown in Figure 2(b), spring-bracket assemblies were used to apply sustained load to beams, which were reinforced with two bars on the tension side. The concrete specimens were demolded after 1 day of cast and then conditioned in water. The beams were loaded right after 7 days of casting. The load level was controlled by the axial stiffness (displacement) of the springs. The tensile strain level of the middle sections of GFRP bars in the loaded beams was in the range of $2000\text{--}2600\text{ }\mu\epsilon$, which corresponds to the sustained stress limit for these GFRP bars suggested by ACI Committee 440 to avoid creep rupture and fatigue fracture.⁸ To monitor the strain level of bars in loaded beams, strain gauges were bonded on the surface of GFRP bars before they were embedded in concrete for two pairs of concrete beams. A data acquisition system was used to record the strain level of bars every half hour. For each pair of concrete beams (with two bars for each beam), the strain levels were monitored up to 2000 h. The strain changes were within 10% of the original levels, and the strain levels approached constant values within 2000 h, indicating that there were nearly no changes in displacement of the springs. Thus, we assumed that almost constant sustained tensile strains in GFRP bars were successfully applied by using the spring-bracket assemblies. In order to investigate the effects of load level on the durability of GFRP bars, a lower load level was applied to two pair of beams to achieve tensile strains of $1100\text{--}1300\text{ }\mu\epsilon$ in GFRP bars. We note that even these sustained tensile strain values were



Figure 1. Glass fiber reinforced polymer (GFRP) specimens.

much higher than what GFRP bars may be subjected to in actual structures designed according to current design guidelines.²¹

Testing plan

In this study, groups of GFRP bars were tested for tensile strength before and after being conditioned for different exposure times, since changes in tensile strength values can reliably be used as indicators of the durability performance.⁴ The test was conducted in accordance with ASTM D 3916-94²² with some modifications. The end-grip used in this study was a steel pipe cut lengthwise into two halves. After being coated with epoxy in their inner surface, the split pipes were bonded on each end of the bar. The tests were carried out on a Baldwin machine. The duration of the loading was 2 to 4 min for each of the specimens.

Temperature exposures of 20, 40, 50, and 60°C were adopted to accelerate the attack of concrete environment on the specimens. These temperatures were well below the glass transition temperature of the resins of selected GFRP bars, which is above 100°C . Concrete beams with embedded bars were immersed in temperature-controlled tap water inside double-walled tanks, which were custom-designed to maintain constant elevated temperature environment as shown in Figure 3(a). Due to space limitation, only GFRP1-bar beams were subjected to controlled temperatures.

For comparison purposes, a few concrete beams and bare GFRP bars were also immersed in ambient tap water of curing tanks as shown in Figure 3(b), or left in the open air of indoor condition. The environmental exposure plan for this study is summarized in Tables 1 and 2. The bars in concrete beams were carefully extracted for tensile tests after conditioning.

In order to investigate the microstructure and chemical composition change of conditioned GFRP bars, samples as shown in Figure 4 were also prepared

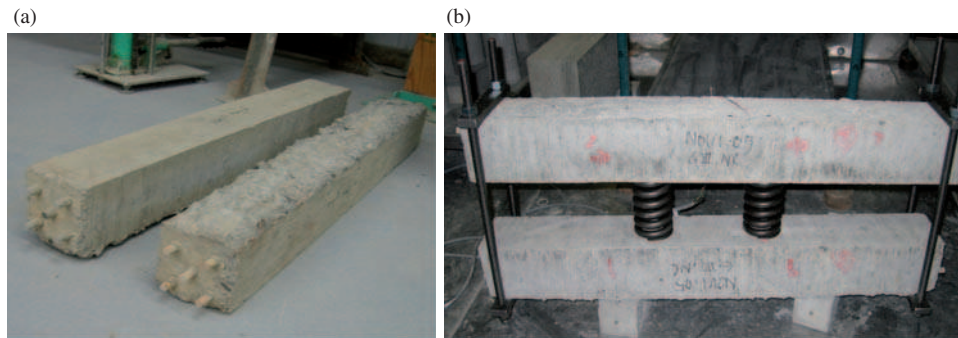


Figure 2. Glass fiber reinforced polymer (GFRP) reinforced concrete beams.

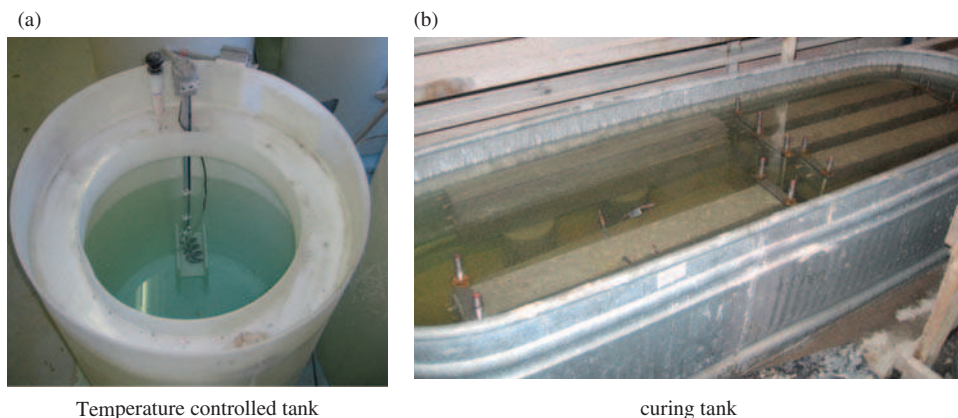


Figure 3. Conditioning tanks.

Table 1. GFRP1 bars specimens in temperature-controlled tanks

Load type	Temperature (°C)	Exposure time (days)				
Not loaded	60	30	90	150	210	270
	50	30	90	150	210	270
	40	30	90	150	210	270
	20	30	90	150	210	270
Loaded	60	90	120	170	210	
	60 ^a	120	210			
	40	90	120	210		
	20	90	120	170	210	

^aLower load level (corresponding to strains of 1100–1300 $\mu\epsilon$) was applied to these specimens.

for SEM. To prepare the samples, a polishing procedure was developed using different grinders and powder and paste materials by trial-and-error. Two samples for each state have been prepared. After polishing, samples were gold-coated for analyses using a Hitachi S-4700 SEM.

Test results

Strength evaluations

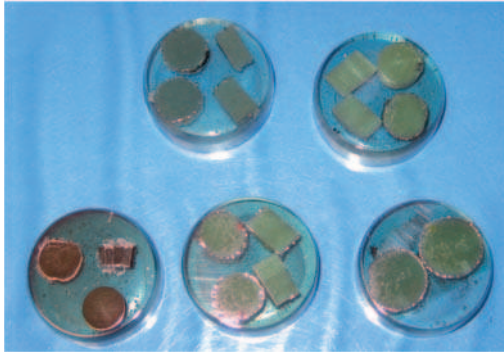
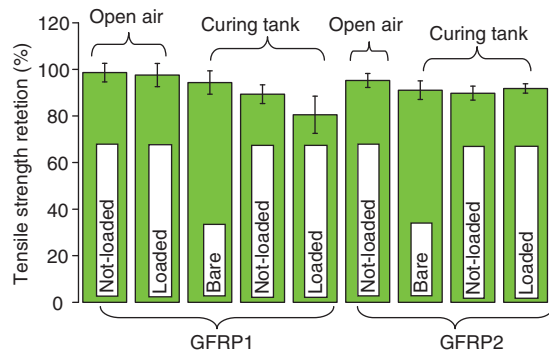
During the tensile strength test, pristine GFRP bars showed approximate linear stress–strain behavior up

to failure, which was characterized by a combination of fiber fracture and delamination. Similar but less catastrophic failure exhibiting more delamination was observed for conditioned specimens. The average tensile strengths and moduli of elasticity for pristine bars were respectively: for GFRP1 856 MPa and 46 GPa and for GFRP2 841 MPa and 49 GPa. The percent tensile strength retentions are summarized in Figures 5 through 7 (the over bar represents standard deviation). Each data represents an average of three to four test results.

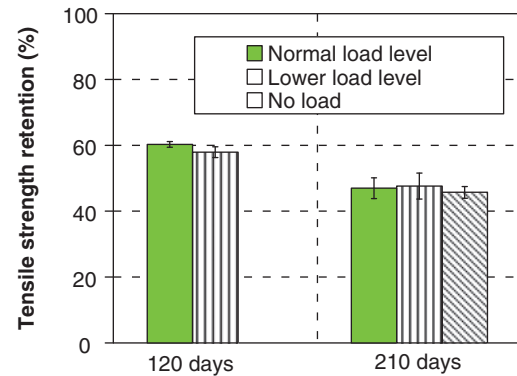
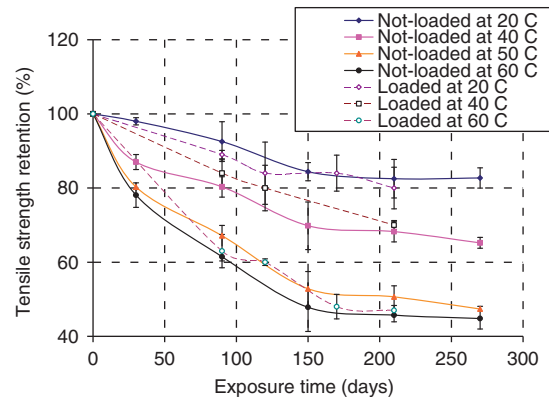
As shown in Figure 5, GFRP1 bars in loaded concrete beams experienced about 20% tensile strength

Table 2. Specimens exposed to room temperature for 150 days

Environmental exposure	GFRP1	GFRP2
Water of curing tank	✓	✓
Within concrete beams in air of concrete lab	✓	✓
Loaded within concrete beams in air of concrete lab	✓	
Within concrete beams in curing tank	✓	✓
Loaded within concrete beams in curing tank	✓	✓

**Figure 4.** Samples for scanning electron microscopy (SEM).**Figure 5.** Tensile strength retention of glass fiber reinforced polymer (GFRP) bars in curing tank or open air of concrete lab after 150 days exposure.

reduction after being immersed in a curing tank for 150 days. For all the other cases, the tensile strength retention was approximately above 90% for both types of bars. There was almost no tensile strength reduction for samples in concrete beams exposed to open air. It was observed that concrete environment (not-loaded concrete beams) is more aggressive to GFRP than direct exposure to water (bare). This may be due to the high alkalinity of concrete environment. Also, GFRP bars had better durability performance in open air of indoor environment than in water (bare) or saturated concrete. GFRP2 bars had relatively better durability performance in concrete environment than GFRP1 bars.

**Figure 6.** Tensile strength retention of GFRP1 bars in concrete beams under different load levels immersed in tap water at 60°C.**Figure 7.** Strength retention of GFRP1 bars in concrete beams exposed to tap water at elevated temperatures.

From Figure 6, it can be observed that the tensile strength retention for GFRP1 bars embedded in beams under both load levels was about 60% after 120 days exposure to water at 60°C. After 210 days exposure, the tensile strength retention was about 45% for GFRP1 bars embedded in beams either loaded or not-loaded. In Figure 7, the tensile strength retentions for GFRP1 bars in not-loaded beams and loaded beam were relatively close to each other. These short-term test results appear to indicate that sustained applied load (relatively low compared to the ultimate tensile strength) in this study did not result in substantially more decrease in tensile strength for GFRP bars in concrete

for up to 210 days. However, maybe due to the long-term effect of sustained loading, the long-term prediction models (by extrapolation of short-term data), presented later in this article, will result in significant prediction differences for not-loaded and loaded conditions. Thus, these initial relative comparisons of discrete values can be misleading and not representative of long-term behavior. From test results shown in Figures 5 to 7, it can be concluded that compared to open-air and immersion in water at ambient temperature, saturated moisture at elevated temperatures within concrete resulted in more severe environmental attack to GFRP bars; while the applied load had relatively less-significant effect.

As shown in Figure 7, the tensile strength decreased with continuous increase of exposure time for GFRP1 bars at all temperatures, and the degradation was more severe for specimens exposed to higher temperatures. But the degradation rate (slope of curves) decreased quickly as the exposure time increased. The same behavior was also found in Ref. 2. The reasons for this phenomenon may be due to factors such as the accumulation of byproducts of chemical reactions, limited replenishment of alkaline solutions within concrete, and decrease in alkalinity (reduced pH value) of pore solution during the degradation process. There is an indication of delayed increase in degradation rates for 210–270 data of not-loaded specimens, though we do not have data after 270 days to support that. But there is a possibility that other degradation mechanisms such as fiber corrosion may start to dominate the deterioration process of GFRP bars and continue to reduce the tensile strength after deterioration rate of fiber/matrix interface approaches zero. More tests need to be carried out to study this.

Microstructure evaluations

The typical results of SEM are presented in Figures 8 and 9. From Figure 8(a) and (b), we can see that the diameter of fibers in GFRP1 is smaller and more uniform (20–25 μm) than that of GFRP2 bars (20–30 μm). Much more embedded polishing material is observed in the matrix of GFRP1 bars than GFRP2 bars. From the analysis of unconditioned specimens, it was found that the properties of matrices and the chemical compositions of fibers may be different, though both bars are composed of E-glass fibers and vinyl ester matrices.

From Figure 8(c) through (f), we observe that many more voids were present in the matrix of GFRP1 bars than of GFRP2 bars, due probably to manufacturing processes. The longitudinal sections of the bars (along the fiber direction as shown in Figure 8(g) and (h)) show that these voids are localized.

For conditioned GFRP1 bars, as shown in Figure 9(a), there is a distinct increase in the number and size of voids, which developed progressively toward the bar-core with exposure time. The longitudinal section of the bar (Figure 9(b)) also shows the increase of number and size of voids along the fiber direction. The same conditions were also observed by.¹¹ A closer view of voids as shown in Figure 9(c) and (d) revealed that the matrix is damaged and the interface is debonded, while the degradation of fibers is not so obvious. For conditioned GFRP2 bars, due to either the short-term exposure or better durability performance of specimens, SEM results did not detect apparent changes. The microstructure comparisons of matrices for the two bars revealed that GFRP2 bars had relatively better durability performance, which agrees with the test results of tensile strength presented above.

It is worth noting that the voids and defects of GFRP bars due to manufacturing and/or degradation are highly localized. This observation supports the argument that the environmental conditioning may degrade significantly the strength but not the stiffness of FRP bars, because the strength is affected by localized weak sections or defects, while the stiffness corresponds to an overall stress–strain characteristic of the bar. Since high magnification images from SEM only captured a limited area of the whole cross section, SEM overall pictures with low magnification were used extensively, and only typical results are presented here.

Durability predictions

In order to predict the durability performance of GFRP bars from short-term data, a prediction model based on knowledge of degradation mechanism should either be developed or adapted from existing formulations. Though the degradation of GFRP bars exposed to water and alkaline solution has been investigated in several studies,²³ there is still no clear understanding of the mechanisms involved.

As discussed in the introductory section, the first degradation model (Equation 4) does not apply to GFRP bars. As an exercise, the test data of this study was used for response prediction based on that model, leading to Arrhenius plots not parallel to each other and invalidating the approach. The second model is based on stress-corrosion of glass fibers, which does not correspond with the observed failure mode in this study characterized by fiber-matrix interface damage, as shown by SEM results. The fiber-matrix interface integrity is critical for load transfer among fibers, and the interface degradation can result in tensile strength reductions due to delamination, as observed in this and other studies (e.g. Ref. 24).

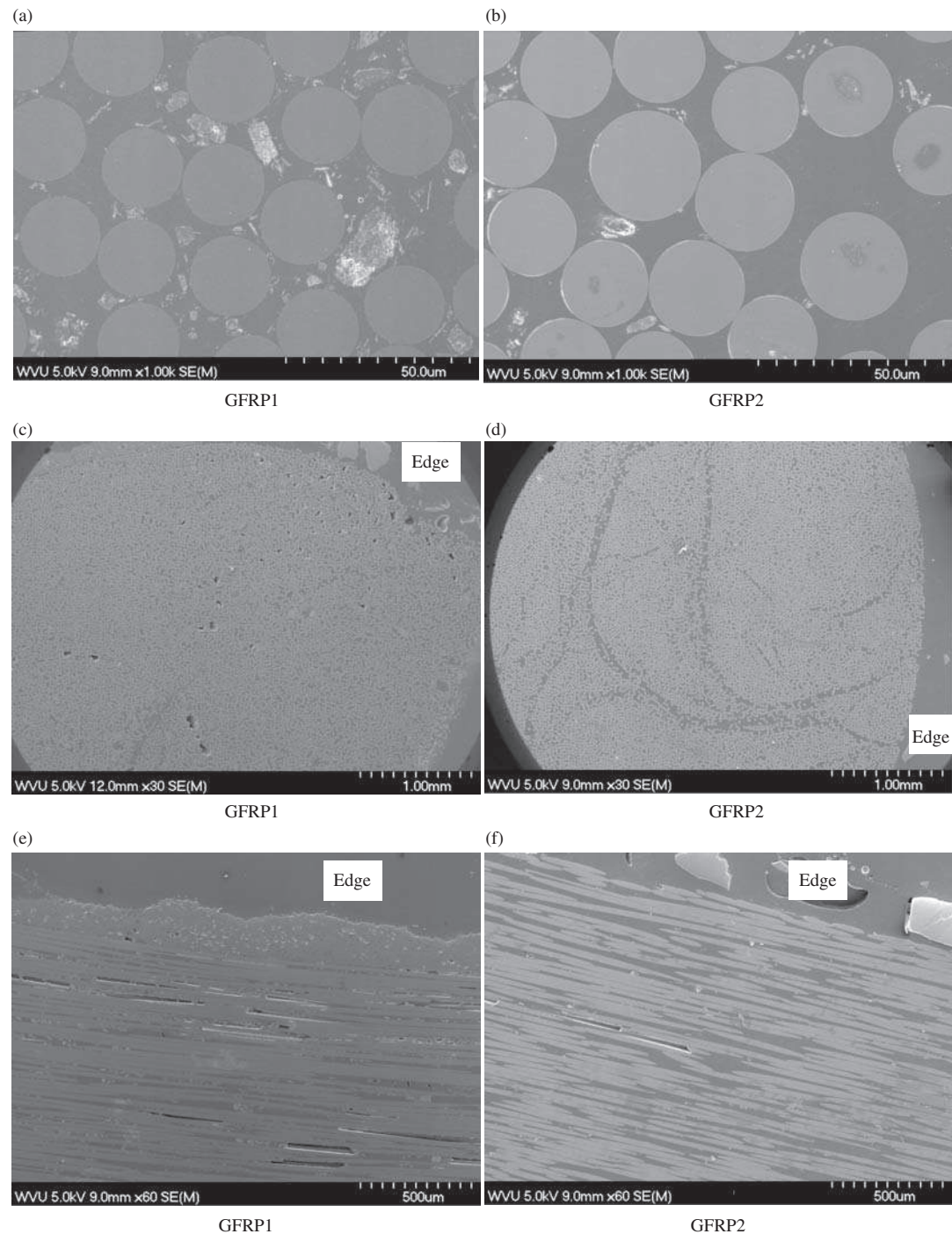


Figure 8. Unconditioned Glass fiber reinforced polymer (GFRP) bars.

In this study, the third and fourth models (Equations 9 and 11, respectively) tried to predict the durability performance of GFRP bars using Arrhenius relation. The short-term test results as shown in Figure 7 were used in the durability predictions, which followed the procedure presented in Refs 4 and 5, and the curve fitting was accomplished using the least square method with the commercial program OriginPro 7.5.

Prediction procedure

For the prediction based on the third model, Equation 9 was first fitted to the test results given in Figure 7 to obtain j and α for different exposure temperatures. As shown in Figure 10, the fitted curves have coefficient of determinations (R^2) above 0.99 for both not-loaded and loaded beams. In the second step, the

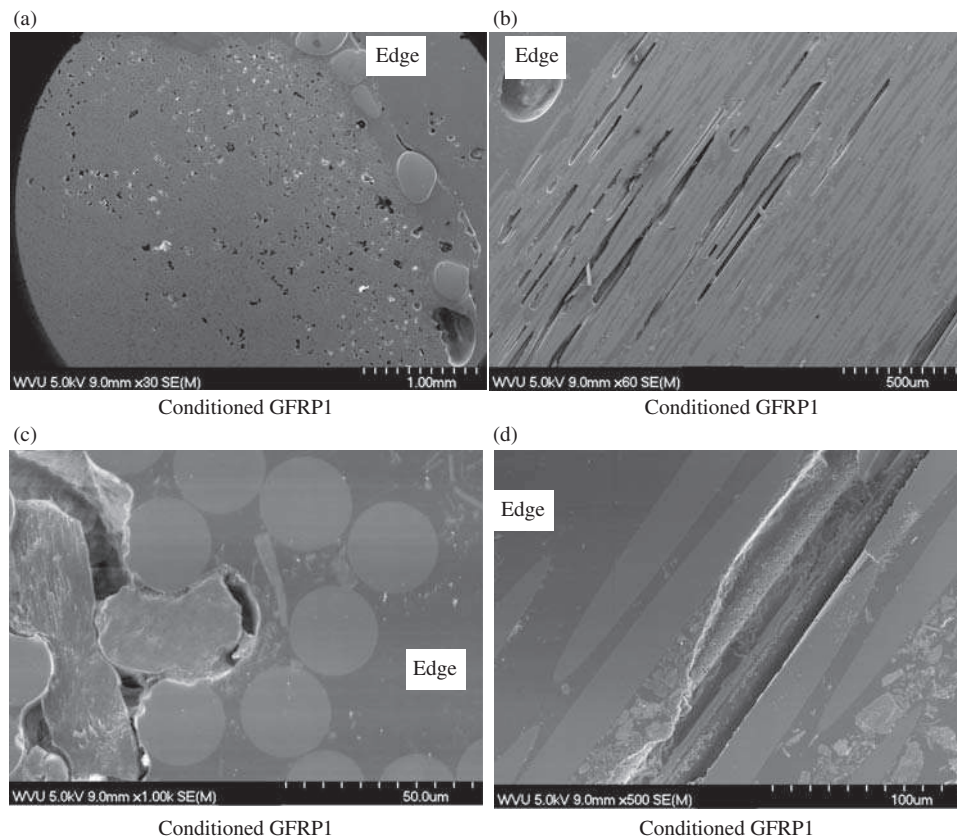


Figure 9. Microstructure of conditioned GFRP1 bars.

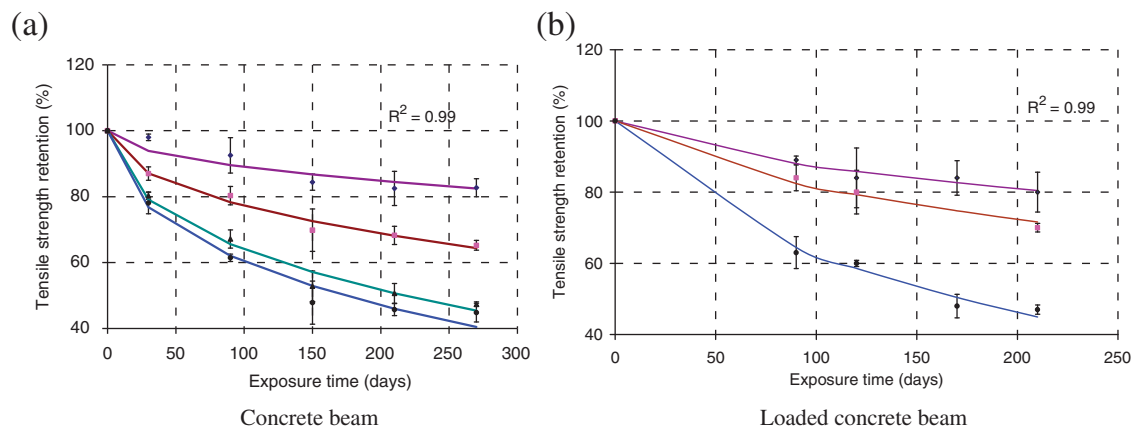


Figure 10. Tensile strength retention vs. time for the third model fitted to the data.

Arrhenius plots, as shown in Figure 11, were built. The times for tensile strength retention to reach 90%, 80%, 70% and 60% at different exposure temperatures were obtained from Equation 9 with regression coefficients j and α obtained in the first step. Equation 3 was fitted to those values as shown in Figure 11. The R^2 for all regression lines were above 0.95 and the straight lines were nearly parallel to each other, indicating that the

accelerated ageing tests were valid and this model may be applied to describe the degradation of GFRP bars in this study. In the third step, from the slopes of straight lines (E_a/R), the activation energy values for not-loaded and loaded specimens were found. Based on the activation energies, the acceleration factors (as shown in Equation 2) for higher temperatures with reference temperature of 20°C could be plotted as shown

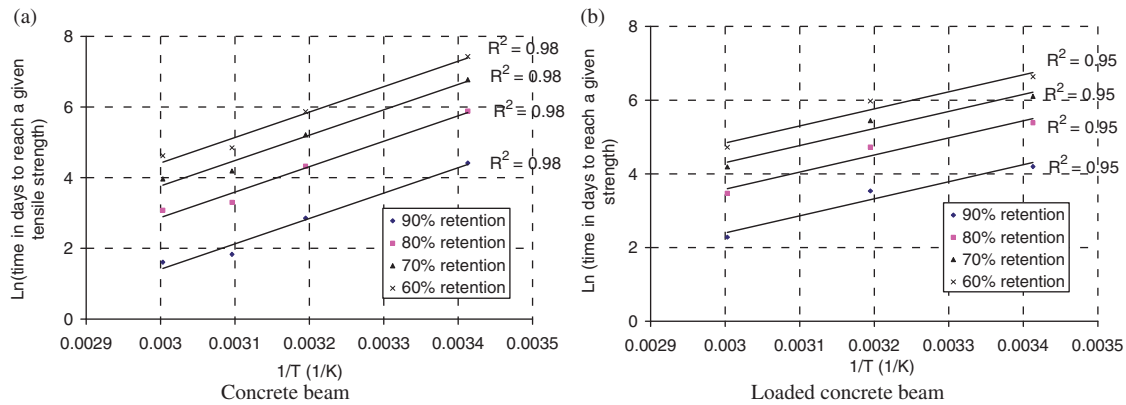


Figure 11. Arrhenius plots of tensile strength retention from the third model.

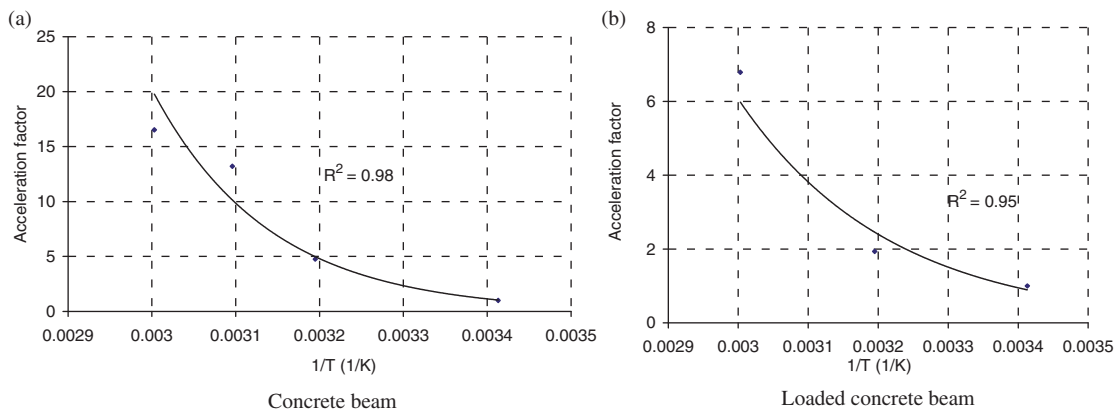


Figure 12. Acceleration factors at reference temperature of 20°C from the third model.

in Figure 12. The acceleration factors were also obtained directly from Equation 9 with regression coefficients j and α obtained in the first step. Those data points were also plotted in Figure 12. In the final step, the exposure times at different temperatures shown in Figure 7 were multiplied with corresponding acceleration factors, and the transformed results are shown in Figure 13. Then by fitting Equation 9 to data in Figure 13, the master curves shown were defined for GFRP1 bars embedded in not-loaded and loaded concrete at 20°C, respectively.

For the prediction using the fourth model, Equation 11 was fitted to the data from Figure 7, with R^2 values above 0.99 as shown in Figure 14. Then, following the same prediction procedure as for the third model described above, Arrhenius plots for this model were built from Figure 14 as shown in Figure 15, where the straight lines with R^2 values above 0.93 were nearly parallel to each other for both not-loaded and loaded beams. This indicated that the accelerated aging tests were valid and this model may also be used to describe the degradation of GFRP bar in this study. Then

acceleration factor curves and master curves were also built for the fourth model prediction as shown in Figures 16 and 17, respectively.

Discussion

The prediction procedures based on the third and fourth models using Arrhenius relations were successfully applied to the test results of this study. It is significant that sustained load, even at such low level of applied tensile stress, accelerated the environmental attack; as noted before, this result was not explicitly apparent from the short-term test results of Figure 7.

In the third model, α is a parameter related to material response to exposure. For the prediction in this study, the α value for not-loaded specimens was found as -0.51 , while for loaded specimens the value was -0.39 . These α values appear to be reasonable based on information in the literature. Prian and Barkatt¹⁷ stated that the dissolution of glass in basic solutions is a constant-rate process, corresponding to $\alpha=0$. It is also noted that in some degradation

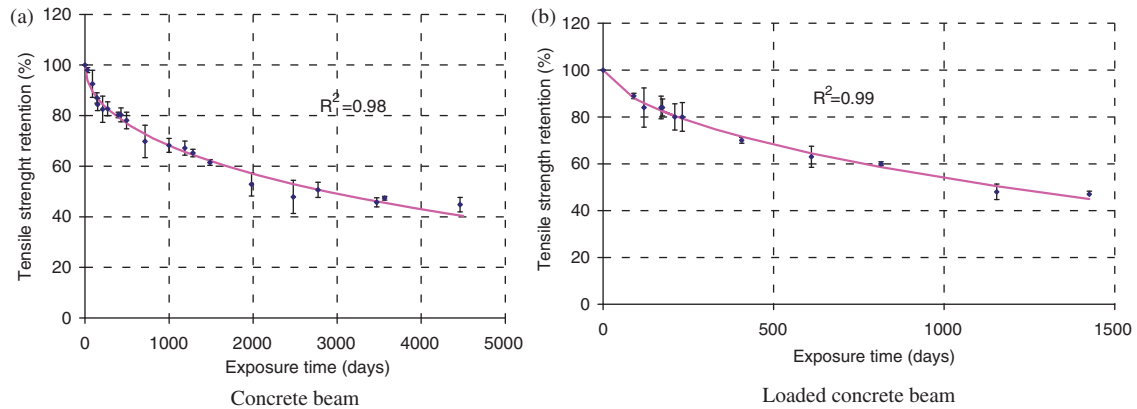


Figure 13. Master curves for GFRP bars embedded in saturated concrete at 20°C from the third model.

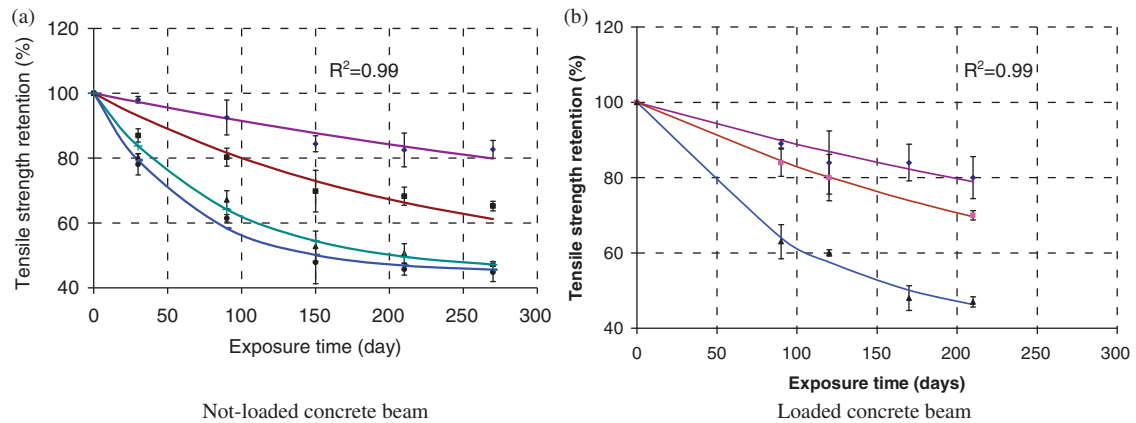


Figure 14. Tensile strength retention vs. time for the fourth model fitted to the data.

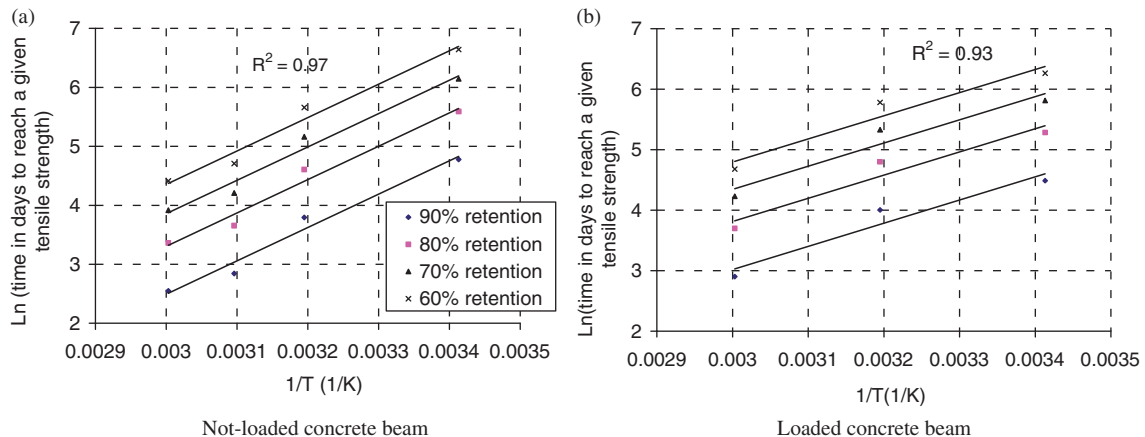


Figure 15. Arrhenius plots of tensile strength retention from the fourth model.

processes, where significant changes of macrostructure of the material occur, such as cracking, the degradation rate has a superlinear relationship with time, so α is greater than zero. In contrast, for degradations where

protective layers are formed, the degradation rate has a sublinear dependence with time, and in this case $\alpha < 0$. For the GFRP bar specimens of this study, and based on test evaluations and results discussed above, the

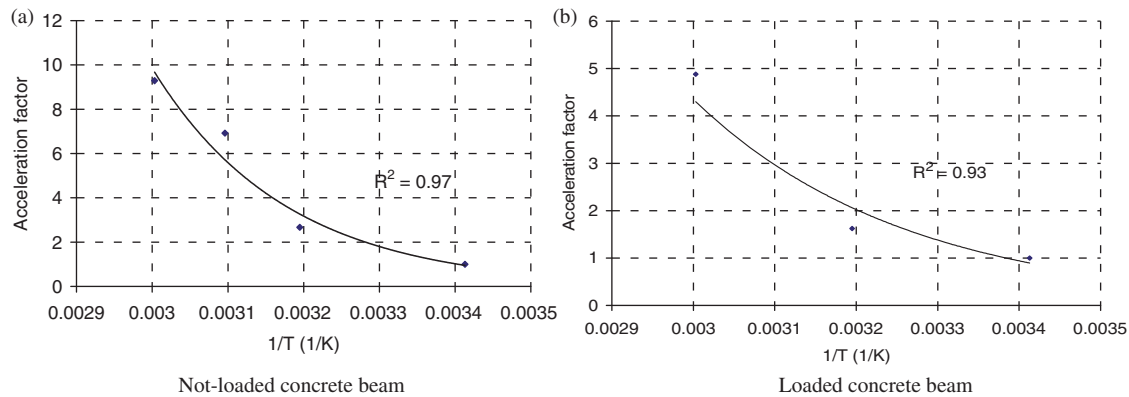


Figure 16. Acceleration factors at reference temperature of $20^\circ C$ from the fourth model.

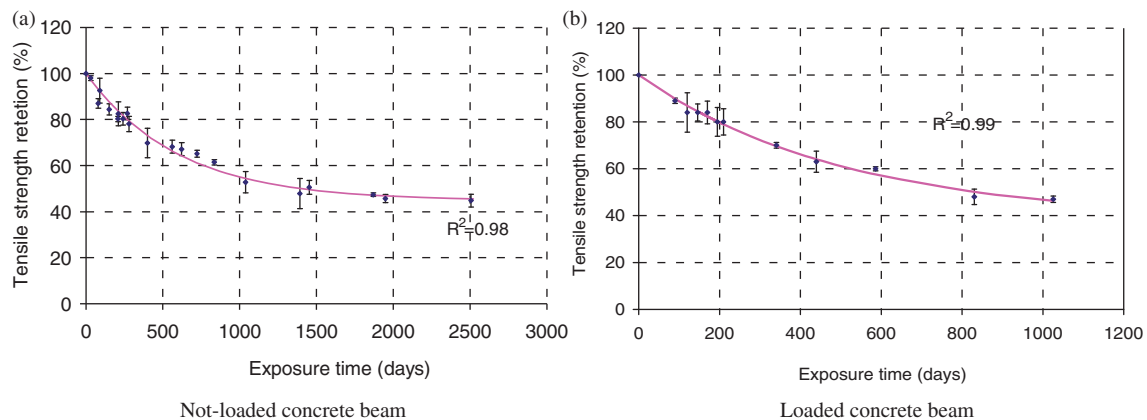


Figure 17. Master curves for GFRPI bars embedded in saturated concrete at $20^\circ C$ from the fourth model.

Table 3. Comparisons of activation energies

Prediction models	Third model		Fourth model		E-V ^a	E-V ^b	E-V ^c	AR-P ^d	AR-C ^e	E-P ^f
	Not- loaded	loaded	Not- loaded	Loaded	Not-loaded					
Activation energy (KJ/mol)	60	39	47	32	41	52	69	39	90	12

^aBare GFRP bars with E-glass fiber and vinyl ester in alkaline solutions (Chen et al.⁴);

^{b,c}GFRP (Gray) and GFRP (FIBERBAR) bars with E-glass fiber and vinylester in saturated concrete cylinders (Dejke²).

^dBare GFRP with AR-glass fiber and polyester in water.

^eAR-glass fiber reinforced concrete in water (Beddow et al.¹⁶).

^fGFRP with E-glass and polyester in water (Phani and Bose¹⁹).

degradation rate decreased rapidly with exposure time as observed in Figure 7. So it is reasonable to expect that the α values obtained from the prediction be less than zero. Since the applied load facilitated the environmental attack by causing concrete cracking and allowing moisture ingress, the α value for loaded specimens was relatively larger (-0.39) than that for not-loaded specimens (-0.51). These α values were close to -0.5 , which indicated that prediction models based

on moisture absorption (as given in Equation (10)) may also be applicable to the test data, especially for not-loaded specimens.

For comparison purpose, the activation energy values obtained in the present durability predictions and results from other studies are listed in Table 3. The types of fibers, matrices, and exposure conditions for other studies are described in the notes below the table. Activation energy is not a measure of durability

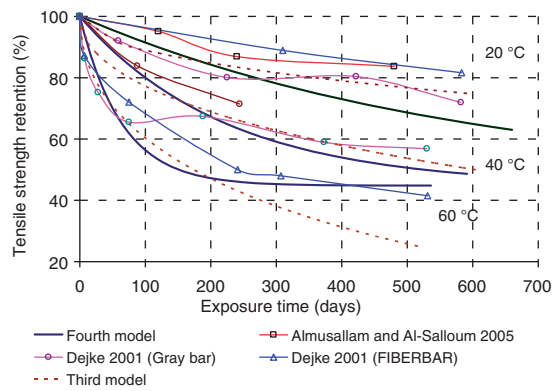


Figure 18. Comparison of predictions and other test results for not-loaded specimens.

but a parameter relating to the temperature and degradation of materials and describing the material response to environmental agents. The activation energy values for loaded specimens are lower than those for not-loaded, since the applied load exacerbated the environmental attack. The activation energy values from this study compare well with those of other studies for E-glass fiber/vinyl ester composites. Since the GFRP bars in Ref. 4 were directly exposed to simulated pore solutions, the degradation in that study was much faster with corresponding lower activation energy. The results of Ref. 2 are comparable to the range of values in this study for not-loaded samples. One can observe from Table 3 that GFRP with polyester matrix had inferior durability performance.

For comparison with existing test results, master curves for different exposure temperatures from both models are produced as shown in Figures 18 and 19. Given the difference of FRP materials, concrete, concrete cover thickness, and loading conditions, the master curves of this study provide closely comparable predictions with others, especially for the loaded specimens. For not-loaded specimens, the proposed master curves can be considered as lower envelope boundaries of tensile strength retentions for the range of results from other studies. This may be due to the aggressive exposure conditions used in this study, since, for example in Ref 2, only the middle portions of the bars were embedded in saturated concrete, and in Almusallam and Al-Salloum,¹² the concrete cover was thicker than that of this study. For loaded specimens, the sustained load in Murkherjee and Arwika¹¹ was about the same as that of this study, while the load applied in Almusallam and Al-Salloum¹² was about twice of the load in this study. The field data of 18 and 30 months from Murkherjee and Arwika¹¹ were obtained from specimens subjected to the exposed hot and humid environment of Bombay, India (the year average temperature is about 27°C). The time predicted to reach 61.4%

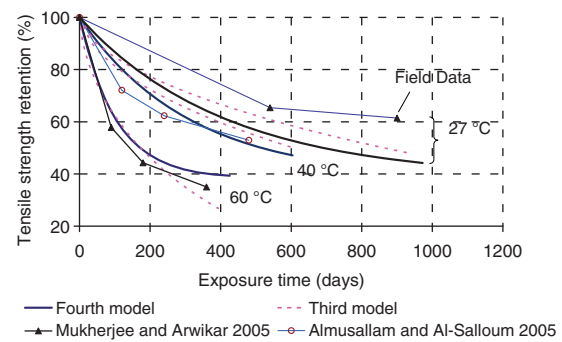


Figure 19. Comparison of predictions and other test results for loaded specimens.

retention from the fourth model master curve at 27°C is 2.2 times sooner than the value reported for ambient conditions of Bombay, India, but the general trend of both predictions and field curves is similar. It is obvious that master curves give conservative predictions compared to available field data, which is expected since natural conditions can generally be less aggressive to GFRP bars due to generally lower temperature and humidity conditions than constant temperature and complete saturation used in the test.

According to Figures 18 and 19, both the third and fourth models account for the decrease of degradation rate with exposure time, but the fourth model predicts that the degradation will tangentially approach a constant state after a period of time, which is also supported by the test results from other studies as shown in Figure 18. As exposure time increases, the predicted tensile strength retention from the third model deviates from the test results and master curves of the fourth model, especially at temperature of 60°C. This discrepancy may also be observed at lower temperatures but for longer exposure times. This deviation may be due to the model limitations for representing the bar degradation response accurately, such as neglecting completely the strength contribution of the affected cross-sectional area. Also, this model may not be able to account for the dominant degradation mechanism of fiber-matrix interface, as supported by the observed SEM images and the failure test results of delamination and fiber fracture.

Thus, based on the above discussion, it is concluded that the fourth model provides more reasonable predictions according to discussions in the previous sections. Also, as corroborated by independent test results (Figure 19), it may be reasonable to assume that the slope of strength vs. exposure time plot for loaded specimens will approach to zero after a certain period of time. But there is a possibility that other degradation mechanisms such as fiber corrosion may start to

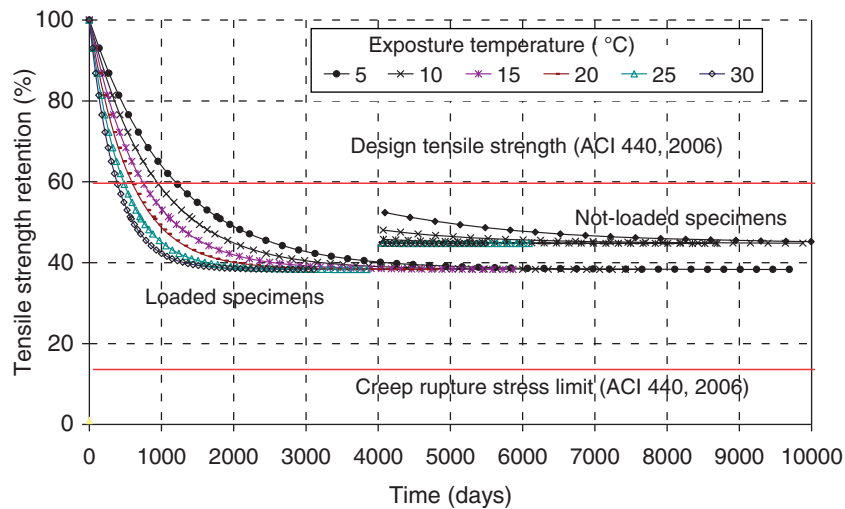


Figure 20. Tensile strength retentions of GFRP bars in saturated concrete.

Table 4 Parameters for master curves of GFRP bars as shown in Figure 17

Temperature (°C)	5	10	15	20	25	30	
Parameters	T (days)						Y_{∞} (%)
Not-loaded concrete beams	2065	1441	1018	728	527	384	45
Loaded concrete beams	1139	893	706	563	451	365	38

dominate the deterioration process of GFRP bars after that. The delayed increase rate observed in Figure 7 and the curvature appeared in Figure 19 may be evidence of the change of degradation mechanism. So it is more reasonable to say that the fourth model is the best to describe the dominant degradation mechanism of GFRP bars of this study.

The tensile strength retentions for GFRP bars in saturated concrete at different exposure temperatures are predicted using the fourth model as shown in Figure 20. The master curves for not-loaded specimens have similar shapes as for loaded specimens, and for clarity of results only the portions of the curves after 4000 days are shown. As shown in Figure 20, degradation rates decrease rapidly and approach to zero. The master curves for different temperatures converge to constant values with the increase of exposure times for specimens either not-loaded, after about 7000 days (19 years), or loaded, after about 4000 days (11 years). The convergence occurs earlier in case of loaded specimens. The values of parameters for those master curves are listed in Table 4, where the long-term retentions (Y_{∞}) are 45% for not loaded and 38% for loaded cases. The design tensile strength and creep rupture stress limit for GFRP bars according to Ref. 8 are also shown in Figure 20; for both limits, an

environmental reduction factor of 0.7 was applied. From this figure, one can conclude that the environmental reduction factor for design tensile strength of GFRP bars from Ref. 8 is unconservative, while the creep rupture stress limit may be too conservative, although this limit may still be applied for long-term and heavily sustained critical load situations; a case that will require further study. However, when considering in-situ conditions, the concrete may not be completely saturated, also the moisture from outside the concrete may not be in direct contact with bars, and the temperature and sustained load may not be constant as the conditioning exposure used in this study. Therefore, the strength reduction may be significantly less than as predicted in Figure 20, which can be adjusted as additional reliable field data for long-term performance becomes available.

Conclusions and recommendations

Based on test results in this study, the following concluding remarks can be made. GFRP2 bars have better durability performance than GFRP1 bars. It is noted that both GFRP1 and GFRP2 bars have been widely used in research for durability of GFRP bars according to the literature. Both of them were called E-glass Vinyl

Ester bars. But few studies investigated both types of bars together. Different and even controversial conclusions were drawn due to the use of those two different types of specimens. It was found that the different durability performance is due to their different composition and/or morphology. The test results and SEM study shed light on the reasons for those controversies. It will also be helpful for future study.

Saturated concrete environment (natural alkaline exposure) is more aggressive to GFRP than direct exposure to water, while GFRP bars in concrete exposed to indoor ambient conditions had better durability performance than bare specimens in water or embedded specimens in saturated concrete. Based on the short-term test results, relatively low load levels applied to specimens in this study showed small effect on degradation. Due to long-term effects of sustained loading, the long-term predictions from resulting master curves showed significant differences between not-loaded and loaded conditions.

According to this study, the dominant degradation mechanism for GFRP bars in saturated concrete was the fiber/matrix interface degradation, which was supported by the SEM images, the tensile test failure mode (a combination of delamination and fiber fracture), and the data analysis using the fourth degradation model in this study.

Based on the experimental results of this study, the tensile strength retention is predicted using the fourth model as 38% for GFRP1 bars after 50-year exposure in saturated and loaded concrete specifically at 10°C, but also valid for a temperature range of 5 to 30°C, due to rapid convergence of prediction curves. Until further data on tensile strength retention are collected from samples in field concrete structures, the current environmental strength reduction factor in ACI 440⁸ can be viewed as unconservative for design tensile strength and conservative for creep rupture stress limit.

According to this study and previous studies,^{4,5} some recommendations can be made for an effective accelerated ageing test method for FRP bars. The tensile strength retention can be used as an indirect measure of the durability performance of FRP bars. FRP bars can be embedded in saturated concrete for conditioning and elevated temperature can be effectively used as acceleration factor. The highest elevated temperature adopted should be lower than the moisture-saturated glass transition temperature (T_g) of the resin, and if this value is not provided, 60°C may be used as the highest elevated temperature. The set of elevated temperatures used in this study such as 20, 40, 50, and 60°C can be a good choice. For the purpose of durability prediction based on Arrhenius relation, test data of at least three different exposure times at three different temperatures (a total of nine) should be collected.

The more data at different temperatures and different exposure times are collected, the more accurate the prediction will be. Also, both exposure temperatures and exposure times should be chosen to be distinctly apart within a given range, in order to obtain apparent degradation results. To improve prediction capability, the exposure time should be long enough, so that the slope of strength retention vs. exposure time plot approaches to zero. Before the prediction functions are developed, the Arrhenius plots should be shown to be straight lines with high correlation coefficients (e.g. $R^2 > 0.8$) and nearly parallel to each other, to check the consistent degradation mechanism induced by accelerated tests.

A large group of different GFRP bars should be studied for long-term durability behavior in order to recommend a general environmental reduction factor for GFRP bars in design guides. The effects of parameters such as humidity of concrete, load level, and concrete cover thickness on the degradation of embedded GFRP bars should be investigated. The dominant degradation mechanism and possible change of degradation mechanism for GFRP bars in accelerated aging and field conditions should be studied. Long-term data of GFRP bars from field conditions should be collected, since correlations are needed between degradations of FRP bars in accelerated testing environment and natural conditions, in order to recommend practical and realistic guidelines.

Acknowledgments

We appreciate the financial support from the Korea Institute of Construction Technology (KICT), Republic of Korea. We thank Doug Gremel of Hughes Brothers and Sam Steere III of Concrete Protection Products for supplying FRP bars and for their valuable technical suggestions.

References

1. Clear KC, Hartt WH, McIntyre J and Lee SK. Performance of Epoxy-coated Reinforcing Steel in Highway Bridges, *NCHRP Report 370*, Transportation Research Board, Washington, DC, 1995.
2. DeJke V. Durability of FRP Reinforcement in Concrete—Literature Review and Experiments, Thesis for the Degree of Licentiate of Engineering, Chalmers University of Technology, Goteborg, Sweden, 2001.
3. Gonenc O. Durability and Service-life Prediction of Concrete Reinforcing Materials, MS Thesis, University of Wisconsin-Madison, Madison, Wis, 2001.
4. Chen Y, Davalos JF, Ray I and Kim HY. Accelerated aging tests for evaluations of durability performance of FRP reinforcing bars for concrete structures. *Composite Structures* 2007; 78(1): 101–111.
5. Chen Y, Davalos JF and Ray I. Durability prediction for GFRP reinforcing bars using short-term data of accelerated aging tests. *J Compos Constr* 2006; 10(4): 279–286.

6. Mufti A, Onofrei M, Benmokrane B, Banthia N, Boulfiza M, Newhook J, et al. Durability of GFRP Reinforced Concrete in Field Structures, *Proceeding of 7th International Symposium on Fiber Reinforced Polymer Reinforcement for Reinforced Concrete Structures (FRPRCS-7)*, Kansas City, MO, 2005, pp. 1361–1377.
7. Japanese Society of Civil Engineering (JSCE). Recommendation for Design and Construction of Concrete Structures Using Continuous Fiber Reinforcing Materials, *Concrete Engineering*, Series 23, JSCE, Tokyo, Japan, 1997.
8. American Concrete Institute (ACI). Guide for the Design and Construction of Concrete Reinforced with FRP Bars. ACI 440.1R-06, *Farmington Hills*, MI, 2006.
9. Bakis CE, Al-Dulaijan SU, Nanni A, Boothby TE and Al-Zahrani MM. Effect of cyclic loading on bond behavior of GFRP rods embedded in concrete beams. *J Compos Technol Res* 1998; 20(1): 29–37.
10. Bank LC, Puterman M and Katz A. The Effect of Material Degradation on Bond Properties of Fiber Reinforced Plastic Reinforcing Bars in Concrete. *ACI Materials Journal* 1998; 95(3): 232–243.
11. Mukherjee A and Arwikaar SJ. Performance of glass fiber-reinforced polymer reinforcing bars in tropical environments -Part I: Structural scale tests. *ACI Structural J* 2005; 102(5): 745–753.
12. Almusallam TH and Al-Salloum Y. Durability of GFRP rebars in concrete beams under sustained loads at severe environments. *J Compos Mater* 2006; 40(7): 623–637.
13. Bank CL, Gentry TR, Thompson BP and Russell SJ. A model specification for FRP composites for civil engineering structures. *Constr Build Mater* 2003; 17(6–7): 405–437.
14. Litherland KL, Oakley DR and Proctor BA. The use of accelerated aging procedures to predict the long term strength of GRC composites. *Cement Concrete Res* 1981; 1(11): 455–466.
15. Caceres A, Jamond RM, Hoffard TA and Malvar LJ. Accelerated testing of fiber reinforced polymer matrix composites-test plan. *SP-2091-SHR*, Naval Facilities Engineering Service Center, Port Hueneme, CA, 2000.
16. Beddow J, Purnell P and Mottram JT. Application of GRC accelerated ageing rationales to pultruded structural GRP. *9th International Conference on Fiber Reinforced Composites*, University of Newcastle upon Tyne, UK, 2002, pp. 215–221.
17. Prian L and Barkatt A. Degradation mechanism of fiber-reinforced plastic and its implications to prediction of long-term behavior. *J Mater Sci* 1999; 34(16): 3977–3989.
18. Katsuki F and Uomoto T. Prediction of deterioration of FRP rods due to alkali attack, *Non-metallic (FRP) Reinforcement for Concrete Structures*, RILEM. London, UK, 1995.
19. Phani KK and Bose NR. Temperature dependence of hydrothermal ageing of CSM-laminate during water immersion. *Compos Sci Technol* 1987; 29(2): 79–87.
20. Chen Y, Davalos JF and Ray I. Life-cycle durability prediction model for GFRP bars under sustained loading and environmental exposure, *Proceeding of 8th International Symposium on Fiber Reinforced Polymer Reinforcement for Concrete Structures (FRPRCS-8)*, University of Patras, Patras, Greece, July 16–18, 2007.
21. El-Salakawy E, Benmokrane B, El-Ragaby A and Nadeau D. Field investigation on the first bridge deck slab reinforced with glass FRP bars constructed in Canada. *J Compos Constr* 2005; 9(6): 470–479.
22. American Society for Testing and Materials (ASTM). *Annual Book of ASTM Standards*, West Conshohocken, PA, 2000.
23. Byars EA, Waldron P, Dejke V, Demis S and Heddadin S. Durability of FRP in concrete – deterioration mechanisms. *Int J Mater Product Technol* 2003; 19(1–2): 28–39.
24. Debaiky AS, Nkurunziza G, Benmokrane B and Cousin P. Residual tensile properties of GFRP reinforcing bars after loading in severe environments. *J Compos Constr* 2006; 10(5): 370–380.